NeuroAgentBench

Paper 3 — A Reproducible Benchmark for Scientific AI Agents in Neuroimaging

Full research-to-publication blueprint • Concept, literature, open-source tool, benchmark protocol, experiments, manuscripts, journal strategy, reproducibility, and adoption

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RESEARCH STATUS | Proposed research program, not completed experiments, validated outcomes, or a submitted manuscript.

Executive decision brief

This project proposes a domain-grounded evaluation framework for AI agents executing neuroimaging tasks. Its distinctive target is not whether an agent can write plausible commands, but whether it produces scientifically correct, reproducible artifacts in controlled execution environments. The public product (NeuroAgentBench) is a task registry, execution sandbox, deterministic scorer, audit trail, and optional results viewer.

Dimension	Initial plan	Submission-level expectation
Research question	How reliably can agents perform neuroimaging computing?	Clearly delimited scientific validity claims
Pilot scope	30–50 expert-reviewed tasks	Expanded representative suite; sample size justified
Task families	BIDS, segmentation, diffusion, connectomics, QC, HPC	Diverse difficulty and held-out cases
Output	Public evaluation runner + paper	Versioned benchmark, metadata, ground truth, replication
Venue strategy	Scientific Data resource vs AI-method paper	Separate manuscript formats, avoid conflation

1. Research concept and significance

1.1 Problem

Scientific AI agents are increasingly able to navigate code, command-line tools, repositories and datasets. Conventional success metrics (plausible explanations, passing software tests, or command completion) do not establish that neuroimaging operations preserve anatomy, coordinate systems, metadata semantics, atlas identities or reproducible provenance. A benchmark should distinguish task completion, software validity, scientific validity, and reproducibility.

1.2 Scope and novelty claim

Proposed contribution: (i) neuroimaging-specific, expert-reviewed realistic tasks; (ii) artifact-level scoring tied to domain constraints and data lineage; (iii) isolated, replayable tool environments; (iv) differential scoring of syntactic, computational and scientific correctness; and (v) documented benchmark-construction and annotation reliability. Novelty must be tested against ScienceAgentBench, SciAgentArena, MLR-Bench,

MLAgentBench, AgentBench, SWE-bench and existing neuroimaging standards. No claim that this is the first scientific-agent benchmark is defensible.

1.3 Primary research questions

1. RQ1: What fraction of neuroimaging engineering tasks are completed correctly by diverse agents under fixed environments and budgets?
2. RQ2: How often do apparently successful executions produce scientifically invalid output?
3. RQ3: Which failure categories dominate by modality, pipeline stage, and task complexity?
4. RQ4: How stable are outcomes across repeated runs, seeds, datasets and agent scaffolds?
5. RQ5: Does access to scientific capability contracts or deterministic semantic validators reduce failure rates? (secondary cross-paper analysis).

1.4 Testable hypotheses (to preregister)

- H1: Apparent execution success exceeds artifact-level scientific success; assess paired differences and confidence intervals.
- H2: Multi-stage and coordinate-sensitive tasks have lower scientific success than metadata-only tasks, conditional on task complexity.
- H3: Semantically enriched tools improve scientific correctness over conventional tool exposure when evaluated with the same agent and budget.
- H4: Repeated agent runs reveal meaningful reliability variance, even on fixed tasks.

2. Literature review and novelty positioning

This section is a targeted scoping review based on verified sources available in October 2026. It is not a systematic or exhaustive search. Before manuscript submission, freeze a formal search strategy, search dates, inclusion criteria and screening spreadsheet.

Prior work	Already established	Gap to evaluate
AgentBench (Liu et al., ICLR 2024)	Agent evaluation across eight interactive environments	Agent/environment methodology; does not provide neuroimaging scientific-validity tests
SWE-bench (Jimenez et al., ICLR 2024)	Real issue-to-code tasks and execution-based validation	Repository-grounded tasks and reproducibility; software tests are not neuroimaging scientific QC
MLAgentBench (Huang et al., ICML 2024)	ML experiment execution and iteration	Interactive experimentation; different scientific domain/output semantics
ScienceAgentBench (Chen et al., ICLR 2025)	102 scientific tasks from peer-reviewed papers; expert validation and code execution	Critical closest comparator; replicate its annotation and evaluation lessons, particularly verified 2026 artifacts
MLR-Bench (Chen et al., NeurIPS 2025)	201 research tasks and a judge for open-ended ML research	Judge design and fabricated-result concerns; not neuroimaging-specific artifact lineage
SciAgentArena (Liu et al., 2026 preprint)	Approximately 200 interactive scientific tasks with stepwise verification	Strong nearest emerging cross-domain work; domain semantic validity remains target to establish empirically
BIDS (Gorgolewski et al., Scientific Data 2016)	Neuroimaging data and metadata standard	Reuse to define validated input fixtures; BIDS conformance is not comprehensive scientific QC

BIDS examples and OpenNeuro	Open standard example structures and public MRI datasets	Source reproducible fixtures and public data; BIDS example raw image placeholders may be empty
Scientific Data submission policy	Dataset-centric Data Descriptor format and technical validation	Appropriate resource route; prohibits new result-driven claims in Data Descriptors

2.1 Claim boundaries

- Do not claim the first benchmark of agents in science; multiple published benchmarks exist.
- Do not confuse BIDS validity with a valid registration, atlas mapping, statistical analysis, or quality-control outcome.
- Do not assert domain-general transfer from a benchmark containing only neuroimaging tasks.
- Do not interpret leaderboard ranking as definitive model capability without repeated trials and budgets.
- Frame the contribution as a new validated evaluation resource and methodology where supported by evidence.

2.2 Formal literature search protocol before submission

Search Google Scholar, PubMed, Semantic Scholar/OpenAlex, ACM Digital Library, IEEE Xplore, arXiv and proceedings through a fixed cutoff. Example Boolean query: ("scientific agent" OR "AI agent benchmark" OR "workflow agent") AND (neuroimaging OR biomedical OR scientific workflow OR reproducibility OR validity). Add targeted queries for BIDS, Nipype, QC metrics, benchmark contamination, evaluation reliability, and agent scaffolds. Screen title/abstract then full text; report inclusion/exclusion counts and novelty matrix. Archive source metadata and search date.

3. NeuroAgentBench public tool: product requirements

3.1 Intended users

Agent researchers, neuroimaging software engineers, pipeline maintainers, students, and academic labs who want to evaluate real agent behavior. Early release should be local-first and CPU-friendly with selected synthetic tasks. Full MRI tasks require separate downloadable fixtures and larger compute budgets.

3.2 Required components

Module	Function	First release
Task registry	Machine-readable objectives, inputs, tags, allowed tools, scoring policy	YAML/JSON schema + versioned packages
Environment builder	Pinned dependencies, resource limits, seeded fixtures	Containerized reference images
Agent adapter	Standard messages/actions, timing, budgets, logging	CLI adapter + reference scaffold
Evaluation engine	Executable tests and scientific-specific validators	Deterministic pass/fail + partial scores
Expert review	Double annotation, adjudication, rationale	Annotation protocol + review sheets
Result store	Tasks, versions, seeds, tool traces,	JSONL/Parquet local outputs

	errors, costs	
Dashboard	Failure distributions, task breakdowns, comparisons	Static optional web report
Submission verifier	Re-run and authenticate external results	Later; no trusted public leaderboard initially

3.3 Example task contract

Task ID: NARB-CONNECT-004; goal: produce a connectome from specified DWI tractogram and a labeled anatomical parcellation. Inputs: anonymized synthetic T1-label image, DWI reference, transforms, manifest. Allowed actions: inspect metadata, transform labels, build matrix, create provenance. Scoring: appropriate transform direction, interpolation for labels, atlas integrity, matrix dimensions, provenance capture. Invalid shortcuts must fail even if matrix file exists. Ground truth allows multiple scientifically equivalent solutions and records numerical tolerances.

3.4 Proposed repository architecture

- neuroagentbench/tasks/{bids,anat,dwi,connectome,qc,hpc}/
- neuroagentbench/schemas/{task,artifact,score}.json
- neuroagentbench/environments/{base,cpu,mri}/
- neuroagentbench/evaluators/{format,execution,science,provenance}.py
- neuroagentbench/agents/{baseline,api_adapter}/
- neuroagentbench/data/manifests/ (data bytes stored externally where large)
- neuroagentbench/results/ (local, excluded from source distribution)
- docs/{getting-started,benchmark-card,limitations,ethics}.md

3.5 Example CLI interface (proposed, not implemented)

neuroagentbench list --suite pilot-v0.1
neuroagentbench run --agent baseline --suite pilot-v0.1 --repeats 3
neuroagentbench evaluate --run results/example.jsonl
neuroagentbench report --results results/ --output report.html

3.6 Release and accessibility

Provide an Apache-2.0 or comparable permissible code license only after checking dependencies and institutional ownership. Each task must declare licensing and dataset provenance. Support CPU-only smoke tests, explicit version tags, reproducible artifacts, issue templates and a minimal benchmark card explaining intended use, leakage risks, known limitations, and inappropriate uses. Public demo data must not contain identifiable patient information.

4. Task taxonomy and benchmark construction

Domain	Representative example	Primary oracle	Proposed pilot tasks
BIDS + metadata	Correct metadata errors without altering valid data	BIDS validator + metadata assertions	8
Anatomical processing	Configure segmentation and check output labels	Command + outputs + image/QC tests	6

Registration + space	Choose correct transforms and resampling	Transform validity + label integrity	8
Diffusion + connectomics	Generate correct connectome from given inputs	Matrix invariants + reference comparisons	8
Scientific QC	Detect obvious segmentation or alignment problem	Expert masks/ratings + deterministic thresholds	8
HPC + reproducibility	Repair scripted Slurm issues and capture provenance	Execution trace + provenance completeness	7

Illustrative pilot target: 45 tasks. A mature benchmark may be larger, but task count must be justified by coverage, expert review capacity and measurement precision. Avoid artificial inflation by trivial variations.

4.1 Task provenance and generation

6. Harvest candidate scenarios from public documentation, reproducible open workflows, verified bug classes and domain-expert cases.
7. Generate synthetic cases for controlled errors; clearly label synthetic versus real-world cases.
8. Write independent reference solutions before evaluating agent outputs.
9. Specify all accepted output representations and scientific tolerances.
10. Separate author from reviewer and adjudicator where practical.
11. Classify tasks by modality, tool count, expected execution cost, error family, and openness of solution.
12. Freeze evaluation and report versions to enable future comparability.

4.2 Annotation protocol

Two expert reviewers independently assess task realism, ambiguity, ground-truth validity, required expertise, scoring completeness and confidentiality. Adjudicate disagreements, document rationales, measure raw agreement and an appropriate agreement coefficient (e.g., Cohen kappa where categories support it). For numerical artifacts, use quantitative expected ranges and prespecified tolerances rather than consensus impressions alone. Reserve reviewer time for recurring benchmark audits.

4.3 Splits and contamination control

Maintain public development tasks, public verified tasks, and a separate held-out test set. If private holdout is used, document governance and release schedule; ensure reviewers can independently audit scoring without exposing benchmark answers during evaluation. Use novel, expert-composed task variants rather than merely copying public tutorials. Report likely model pretraining contamination explicitly and maintain semantic-equivalent paired cases.

5. Public datasets and experimental fixtures

Asset	Use	Caution / verification
BIDS examples	Metadata and directory-fixture tests	Many example raw files are empty placeholders
OpenNeuro snapshots	Public, versioned imaging test data	Pin accession, snapshot, selected participants, license and files
Generated NIfTI phantoms	Exact coordinate-space and interpolation oracles	Clearly distinguish from biologically plausible real data
Synthetic transforms and atlases	Controlled connectivity failure injection	Validate transform convention and

		labels
Small reference workflow outputs	Fast artifact tests independent of heavy processing	Check legal redistribution and version pinning
Optional institution datasets	External validation only	Required approvals; not needed for base benchmark

5.1 Dataset selection checklist

- Selected OpenNeuro accessions must be verified against the actual snapshot, modalities, file sizes, license, preprocessing needs and public download status.
- Avoid relying on one site or acquisition protocol if claiming robustness; select heterogeneous available datasets.
- Record checksums for exact input files and archive any synthetic-data generator with seeds.
- Set compute-aware task subsets: metadata-only, lightweight NIfTI, and heavy full pipelines.
- Prohibit PHI and patient-identifying metadata in public benchmark traces.

6. Evaluation design and baselines

6.1 Controlled experimental conditions

Baseline	What it isolates
B0: deterministic reference workflow	Ceiling target for execution and artifact correctness
B1: LLM agent + documentation + shell	General agent abilities with unstructured guidance
B2: agent + typed API/MCP tools	Effect of ordinary programmatic interfaces
B3: agent + scientific capability contracts	Effect of semantic tool metadata
B4: agent + semantic validator	Effect of explicit scientific checking
B5: non-agent scripted baseline	Distinguish agent value from fixed scripted automation

B3 and B4 overlap with Papers 1 and 2; they are secondary comparator conditions and must not become the only novelty claim of Paper 3. Fix model version, scaffolding, prompts, allowed actions, token budget, wall-clock cutoff, execution resources and retry policy. Specify where exact parity is impossible.

6.2 Primary outcome

Primary metric: task-level scientifically valid artifact success rate, binary with prespecified rules for each task. Report numerator, denominator, 95% confidence interval, and counts of infra-failed/censored attempts separately. If a task has no valid automatic oracle, require blinded expert adjudication and report agreement.

6.3 Secondary outcomes

- Executable completion rate; valid file format rate; semantic constraint satisfaction; provenance completeness.
- Error category: incorrect tool, invalid parameters, coordinate mismatch, missing inputs, hallucinated paths, false success claims, incorrect QC interpretation.
- Partial credit with explicitly defined weights; avoid treating arbitrary composite percentages as a single objective truth.

- Resource use: runtime, CPU hours, agent token use, hosted-model cost under recorded pricing, peak memory.
- Reliability: variance and probability of success over repeated seeds/runs; self-repair outcomes.
- Robustness: distribution shift across datasets, unseen tools and task variants.
- Human usefulness (optional): blinded expert comparison of reports and required interventions.

6.4 Scientific artifact validity rubric

Layer	Illustrative checks
L0 — Instruction compliance	Correct task selected; no unauthorized operations
L1 — Syntax and data format	Required files exist; file format and metadata parse
L2 — Computational execution	Reproducible commands complete without hidden exceptions
L3 — Scientific semantics	Coordinate spaces, transformations, atlas, units and modality valid
L4 — Artifact and QC	Numerical invariants, relevant visual QC, expected anatomy, uncertainty
L5 — Reproducibility	Pinned inputs, environment, parameters, logs and provenance

Each layer is scored separately and scientific-valid success is explicitly defined in the task manifest; not all tasks require all layers. A matrix with the expected dimensions but wrong atlas alignment does not receive full credit.

7. Statistical analysis plan

Unit of inference: task, accounting for correlated repeated runs within tasks and shared datasets. The manuscript should prespecify primary comparisons, model/scaffold factors, stratification and handling of execution infrastructure failures. Use a pilot to inform precision and power targets; do not claim an arbitrary task count guarantees statistical adequacy.

- Primary: estimate scientifically valid completion proportions and intervals; use hierarchical logistic regression or clustered bootstrap for task-by-agent repeated observations where appropriate.
- Paired comparisons: evaluate agents on the identical fixed tasks and analyze paired success differences; avoid naive independent-proportion tests.
- Failure modes: present distributions with denominators; exploratory multiplicity-adjusted comparisons if many classes.
- Annotation: report agreement, adjudication counts and changes to the ground truth.
- Robustness: rerun task subset with independent seeds or agent temperature settings and inspect outcome instability.
- Cost: compute cost per scientifically valid completion, including non-successes.
- Sensitivity: change score tolerances and exclude ambiguous tasks; report impact transparently.

7.1 Minimum reproducibility specification

Freeze package SHA, dataset snapshots and checksums, container digests, software licenses, model identifiers and availability date, inference parameters, prompts/system policies, scaffold commit, agent action traces, benchmark evaluator version, random seeds, run timeouts and hardware details. Publish raw task-level outcomes rather than leaderboard totals alone.

8. Threats to validity and mitigation

Threat	Mitigation
Evaluator false negatives	Multiple valid solutions, manual adjudication, evaluator regression tests
Evaluator false positives	Domain-invariant checks, expert audit of passed outputs
Task contamination	Novel tasks, revised variants, disclosures and periodic audit
Tool/scaffold confounding	Fixed interfaces, model cross-over design and ablations
Synthetic realism gap	Mixed synthetic and real public-data examples; stratified reporting
Limited modality coverage	Define scope and hold out cross-domain claims
Leaderboard overfitting	Version splits, locked benchmark sets, submission audit
Compute failures	Separate infrastructure failure from agent error; rerun policy
Clinical misuse	Research-only limitation; no clinical diagnostic claims
Proprietary model drift	Version, date, settings, repeat evaluation

9. Figures, tables, and deliverables for the manuscript

13. Figure 1: Conceptual distinction between command success and scientific artifact validity.
14. Figure 2: Benchmark architecture and isolated task environment.
15. Figure 3: Task distribution by modality, challenge type and difficulty.
16. Figure 4: Performance by agent under the layered validity rubric (with intervals).
17. Figure 5: Error taxonomy and failure analysis; representative anonymized examples.
18. Figure 6: Reliability and generalization across runs, datasets and scaffolds.
19. Table 1: Comparison with ScienceAgentBench, SciAgentArena, MLR-Bench and other prior benchmarks.
20. Table 2: Dataset and task provenance with licensing and expected compute.
21. Table 3: Primary and secondary benchmark results with precision estimates.
22. Supplementary: tasks, validator evidence, annotation rubric, prompts, hyperparameters, ablations and sensitivity analyses.

10. Two different manuscript routes — do not conflate

10.1 Route A: Dataset/resource paper (Scientific Data)

Scientific Data Data Descriptor structure: Background & Summary; Methods; Data Records; Technical Validation; Usage Notes; Data Availability; Code Availability and required statements. The manuscript documents benchmark construction, content, annotation, technical checks, accessibility and reuse. It must not present new hypothesis-testing results or broad substantive AI findings inside the Data Descriptor. Separate agent-performance analyses belong in a distinct research Article or another venue. Editorial criteria prioritize technical quality and reuse, not novelty of scientific findings.

10.2 Route B: AI benchmarking/methodology research paper

Proposed standard structure: Introduction; Related Work; Benchmark Design; Scientific Validity Metrics; Experimental Setup; Results; Failure Analysis; Limitations & Ethics; Reproducibility; Conclusion. A novel, validated evaluation methodology and empirical findings could support Nature Machine Intelligence or a

leading AI-conference datasets/benchmarks venue, but strong cross-system conclusions and wider impact are essential.

10.3 Draft abstract scaffold — research Article (no fabricated findings)

AI agents are increasingly used to automate scientific computation, yet conventional execution metrics may fail to capture domain-specific errors. We introduce NeuroAgentBench, a reproducible benchmark for assessing AI agents on neuroimaging engineering tasks using artifact-level scientific validation. Tasks cover [verified domains] and are constructed from [documented sources] with [expert review procedure]. We evaluate [agents and conditions] using a layered scoring framework separating executable completion, semantic validity, and reproducibility. Our experiments assess [prespecified primary outcomes], failure modes and cross-condition reliability. Results [insert verified quantitative findings only]. The benchmark, evaluation code and documented task resources are [release status].

11. Journal-tier strategy and requirements

Tier / venue	Why it could fit	What is needed
Elite stretch: Nature Machine Intelligence	Broad insight into reliability and evaluation of scientific agents	Conceptually new benchmark/evaluation methodology, multi-agent validation, surprising generalizable findings, strong independent utility
High target: Nature Communications	Important, widely relevant scientific/AI contribution	Robust experimental discovery relevant beyond niche workflow debugging; compelling validation and rigorous analysis
High resource target: Scientific Data	Reusable benchmark dataset and documented ground truth	Repository deposit, thorough construction methods, technical validation, access/licenses, no novel experimental claims in Data Descriptor
Top AI-conference resource track	Machine learning benchmark community adoption	Strong benchmark quality, contamination control, scorer reliability, accessible code/data; venue-specific active track rules
Specialist: GigaScience	Open reproducible computational data/science resource	Substantive open infrastructure, tested workflow, accessible data/code and scientific use
Specialist: Imaging Neuroscience	Neuroimaging methods/software and community utility	Substantial neuroimaging evaluation methodology and validated imaging tasks
Software fallback: JOSS / SoftwareX	Reusable evaluation software	Public development, quality tests, documentation, research utility; papers focused on software rather than AI results

11.1 Highest-tier gate

For Nature Machine Intelligence, a simple domain benchmark is not sufficient. Establish a broadly consequential conclusion: e.g., conventional agent leaderboards substantially overestimate scientific reproducibility under independent validity checks; demonstrate that across heterogeneous agents, task families, and ideally at least one independent scientific field. Identify which components are algorithmically

new and distinguish empirical observation from causal interpretation. Editorial fit and acceptance remain uncertain.

11.2 Scientific Data-specific gate

Deposit benchmark data in a suitable durable repository accessible to anonymous reviewers, document each record/column/task schema, supply quality-control methods, explain source rights, provide usage notes and executable scorer, include data and code availability statements. Keep new agent-comparison hypothesis testing out of the Data Descriptor manuscript; use a paired research publication if results justify it. Confirm word/figure/format requirements at submission time.

11.3 Venue decision tree

If the most important output is a curated, reusable dataset with rigorous technical validation → Scientific Data. If the main advance is an evaluation methodology that changes conclusions about agent capabilities across settings → Nature Machine Intelligence or a major AI benchmark track. If the result is a reliable, useful package without substantial methodological novelty → JOSS/SoftwareX. If neuroimaging-specific methodological impact dominates → Imaging Neuroscience.

12. Project plan and implementation milestones

Phase	Deliverable	Decision gate
0 — Weeks 1–3	Search/novelty matrix; collaborators; ownership checks; protocol	Scope and gap endorsed by independent experts
1 — Weeks 4–6	Task schema, CLI skeleton, 10 synthetic tasks	Reproducible environments and ground truth
2 — Weeks 7–10	45-task pilot and independent review	Low ambiguity; evaluator tests passing
3 — Months 3–5	Agent baselines, repeated pilot runs, scoring audit	Feasible cost and credible discriminating power
4 — Months 5–8	Expanded benchmark and holdout; generalization checks	Primary outcome estimable with defensible precision
5 — Months 8–10	Public benchmark release, replication, analysis	Independent reproduction on clean machine
6 — Months 10–12+	Target-tailored manuscript and submission	Claims supported and required journal format met

Timeline is an illustrative project plan, not a guarantee; expert annotation, compute and institutional approvals can materially change it.

13. People, infrastructure, and budget planning

- Research engineering: benchmark framework, task/environment versions, evaluation runner, automated tests.
- Neuroimaging expert (at least one, preferably multiple independent reviewers): domain validity and QC annotation.
- AI agent evaluation collaborator: baselines, model/scaffold selection, contamination controls and fairness.
- Statistician/methodologist: primary outcome, clustered analyses, uncertainty, reproducibility.

- Resources: workstation or approved HPC, container runtime, public MRI, storage for logs/artifacts, model API/local inference budget.
- Create a cost sheet with per-task CPU/GPU hours, repeated-runs multiplier, human annotation hours and API costs; cap the pilot before scaling.
- Check employer IP, research grants, dataset licensing and any patient-data governance before public release.

14. Public adoption and ethical research collection

Release a useful free local benchmark runner before publication. Public users can benchmark their agents and report failed/scorer-ambiguous tasks. Public usage metrics (installations, integrations, reports) establish ecosystem utility, not causal evidence of scientific accuracy. Collect telemetry only with explicit opt-in and appropriate disclosure. Hold benchmark versions constant for the scientific comparisons; release later versions with migration notes and separate leaderboards. Public submissions require isolated execution, resource quotas and verification before displaying rankings.

15. Critical reviewer objections and planned responses

Likely objection	Required response
"Another benchmark; what is new?"	Formal novelty matrix and empirically distinct scientific-artifact scoring
"Your reference answers are too narrow"	Accept equivalent valid workflows; audit false negatives
"Human QC is subjective"	Prespecified rubric, multiple raters, adjudication and uncertainty
"Synthetic tasks are unrealistic"	Public-data tasks and documented real-world scenario provenance
"Agent rankings reflect scaffolds rather than models"	Factorial/cross-over testing and detailed scaffold reporting
"Not generalizable outside neuroimaging"	Limit claims; add cross-domain validation only if done
"No genuine independent reproduction"	Release single-command evaluator and independent replication log
"Scientific Data is not the right venue for benchmark results"	Separate Data Descriptor from hypothesis-testing research Article

16. Required manuscript and release package

Before testing

- [ ] Frozen protocol and hypotheses
- [ ] Task-generation policy and annotation rubric
- [ ] Dataset snapshots/licenses and data inventory
- [ ] Reference solutions and validator unit tests
- [ ] Agent model/scaffold versions and resource budget

Before journal submission

- [ ] Complete and auditable task manifests
- [ ] Expert sign-off and annotation agreement
- [ ] Results by task and agent with uncertainty

- [ ] Robustness, ablations and failure analysis
- [ ] Independent clean-room run on separate machine
- [ ] Public code/data or legally justified restricted access
- [ ] Journal-specific manuscript, cover letter and disclosure statements

17. Immediate first implementation sprint

23. Create repo with task.schema.json, task manifest examples and benchmark card.
24. Build three deterministic synthetic tasks: missing BIDS field, wrong-space transform, mislabeled atlas matrix.
25. Create validators independently from agent-generated solutions.
26. Implement a containerized, read-only task runner that records agent actions and resource use.
27. Recruit two experts to review task ambiguity and scientific scoring.
28. Run two agent setups against the same 10 pilot tasks with three repeats.
29. Hold a protocol review meeting; decide whether to expand task scope or refine the scorer.

18. Core references and actionable sources

Links below lead to primary papers, proceedings, or official documentation. Exact bibliographic metadata should be exported into a citation manager and checked before manuscript submission.

[1] AgentBench (Liu et al., ICLR 2024). https://proceedings.iclr.cc/paper_files/paper/2024/hash/e9df36b21ff4ee211a8b71ee8b7e9f57-Abstract-Conference.html

[2] SWE-bench (Jimenez et al., ICLR 2024). https://proceedings.iclr.cc/paper_files/paper/2024/hash/edac78c3e300629acfe6cbe9ca88fb84-Abstract-Conference.html

[3] MLAgentBench (Huang et al., ICML 2024). https://proceedings.mlr.press/v235/huang24y.html

[4] ScienceAgentBench (Chen et al., ICLR 2025). https://hal.cs.princeton.edu/scienceagentbench

[5] MLR-Bench (Chen et al., NeurIPS 2025). https://proceedings.neurips.cc/paper_files/paper/2025/hash/ab8dd000d6f87f40061a73f8bca7fae4-Abstract-Datasets_and_Benchmarks_Track.html

[6] SciAgentArena (Liu et al., 2026 preprint). https://arxiv.org/abs/2606.12736

[7] BIDS (Gorgolewski et al., Scientific Data 2016). https://www.nature.com/articles/sdata201644

[8] BIDS examples and OpenNeuro. https://bids.neuroimaging.io/datasets/examples.html

[9] Scientific Data submission policy. https://www.nature.com/sdata/submission-guidelines

[10] MLAgentBench (Huang et al., ICML 2024). https://proceedings.mlr.press/v235/huang24y.html

[11] SWE-bench (Jimenez et al., ICLR 2024). https://proceedings.iclr.cc/paper_files/paper/2024/hash/edac78c3e300629acfe6cbe9ca88fb84-Abstract-Conference.html

[12] BIDS Standard documentation. https://bids.neuroimaging.io/

[13] OpenNeuro user guide. https://docs.openneuro.org/user-guide/

[14] SciAgentArena repository. https://sciagentarena.github.io/

[15] ScienceAgentBench implementation and 2026 verified split. https://github.com/OSU-NLP-Group/ScienceAgentBench

[16] Nature Machine Intelligence aims & scope. https://www.nature.com/natmachintell/aims

[17] Scientific Data peer review policy. https://www.nature.com/sdata/policies/editorial-and-publishing-policies

[18] NeurIPS 2025 MLR-Bench proceedings. https://proceedings.neurips.cc/paper_files/paper/2025/hash/ab8dd000d6f87f40061a73f8bca7fae4-Abstract-Datasets_and_Benchmarks_Track.html

Final recommendation

Build NeuroAgentBench as a public, versioned evaluation resource, but keep the scientific claim sharply focused: artifact-level validity and repeatability of neuroimaging agents, beyond task completion or code execution. Target Scientific Data for a rigorously annotated reusable dataset, or a higher AI-methods venue only when experiments show a novel and generalizable finding. Share an evaluator that survives independent reproduction, not merely a leaderboard.